

1. Power Architecture

The system utilizes a **multi-voltage architecture** derived from a main 48V traction battery. All subsystems share a **Common Ground (GND)** reference to ensure signal integrity between high-voltage LEDs and low-voltage logic.

- **Source:** 48V DC Battery.
- **12V Rail (via Buck Converter):** Powers the **WS2811 Addressable LED Strips** (Headlights, Taillights, Underglow).
- **5V Rail (via Buck Converter):** Powers the **Raspberry Pi 4, 8x Ultrasonic Sensors, and Relay Module.**
- **3.3V Rail (via Raspberry Pi):** Powers sensitive logic modules (**NFC Reader, DHT11, Buzzer**).

2. Input Subsystems (Sensors)

- **Parking Assist (Ultrasonic Array):**
 - **Hardware:** 8x HC-SR04 Sensors (4 Front, 4 Rear).
 - **Logic:** Uses a **Shared Trigger** configuration (One GPIO triggers all sensors simultaneously) to save pins.
 - **Safety Circuit:** Each "Echo" pin utilizes a **Voltage Divider ($1k\Omega + 2k\Omega$)** to step down the 5V return signal to a safe 3.3V for the Raspberry Pi.
- **Access Control:**
 - **Hardware:** RC522 NFC/RFID Reader.
 - **Interface:** SPI (Serial Peripheral Interface).
 - **Function:** Authenticates user key cards to unlock the motor relays and dashboard.
- **Climate Monitoring:**
 - **Hardware:** DHT11 Sensor.
 - **Function:** Digital Temperature and Humidity reading.

3. Output Subsystems (Actuators)

- **Smart Lighting:**
 - **Hardware:** WS2811 Addressable LED Strip (12V).
 - **Topology:** Single data line (GPIO 18) daisy-chained to control Front, Interior, and Rear zones independently via software.
- **Motor Control:**
 - **Hardware:** 2-Channel 5V Relay Module.
 - **Function:** Physically cuts or enables the ignition/motor circuit based on NFC authentication.
- **Audio Warning:**
 - **Hardware:** Active Piezo Buzzer.
 - **Function:** Provides acoustic feedback for parking proximity and system alerts.

4. Software Architecture

- **OS:** Raspberry Pi OS (Bookworm/Debian 12).
- **Interface:** Python 3 + **PyQt5** (Touchscreen GUI).
- **Back-End:**
 - **hardware.py:** Handles GPIO interrupts, threading for NFC/Sensors, and LED signal generation (`rpi_ws281x`).
 - **main.py:** Manages the GUI, State Machine (Locked/Unlocked), and visualizes sensor data.